

# Exact Molecular Docking: A Machine-Checked Theory of Configuration Resolution, Complexity, and Thermodynamic Cost

Tristan Simas  
McGill University  
`tristan.simas@mail.mcgill.ca`

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## Abstract

Exact molecular docking is treated as exact configuration resolution under constraints. The abstract theory is stated for bounded decision systems through sufficient coordinate sets, structural rank, and decision-quotient entropy. Constrained molecular systems instantiate the same framework through holonomic constraints, cutoff-local interaction structure, and sampled action families.

General exact sufficiency certification contains a hardness core, while cutoff locality bounds docking structural rank by active-site and ligand coordinates and therefore isolates a theorem-backed low-rank regime. For sampled docking, exact and coarse winner sets agree under an explicit half-gap hypothesis, and the retained inside-cutoff coordinates are sufficient under the stated compatibility assumptions.

The same exact-resolution structure carries irreducible thermodynamic cost once a positive per-bit lower bound is fixed. Under Landauer calibration, any exact-resolution cycle satisfies

$$E \geq k_B T H_{\text{nats}}(D),$$

where  $H_{\text{nats}}(D)$  is the natural-log entropy of the decision quotient. In the canonical binary encoding, this bound sharpens to

$$E \geq \text{DOF}(A) k_B T \ln 2,$$

and the rank-1 regime is the unique thermodynamic ground state: every system with more than one degree of freedom lies strictly above the minimum per-cycle resolution cost.

The structural part of the argument is finite. Bounded systems admit only finitely many acquisition events. Exact resolution requires a sufficient coordinate set. The associated canonical decision problem records one binary acquisition channel for each degree of freedom, and the number of independent coordinates is identified exactly with the structural rank of the encoded decision problem:

$$\text{DOF}(A) = \text{srnk}(\text{canonicalDP}(A)).$$

The decision quotient therefore has at most  $2^{\text{DOF}(A)}$  optimal-action classes, so its entropy is controlled by the same coordinate count that governs exact physical resolution.

For constrained molecular systems with  $N$  atoms and  $k$  independent holonomic constraints, the exact-resolution floor scales with  $3N - k$ . Thresholded one-bit readout channels and two-level atomic transition systems instantiate the same binary interface.

For bounded regions this yields the bounded-acquisition inequality

$$\text{DOF}(A) \leq \frac{c\tau}{d},$$

and corresponding bounds on decision-class count from spacetime and energy budget.

Theorem-level mismatch and residual witnesses place nonideal implementations strictly above the Landauer floor. An explicit binary mismatch witness and an explicit two-state residual witness each yield at least one additional per-bit lower-bound unit above the Landauer floor and raise the energy-information coefficient above  $k_B T$ . Finite-capacity substrates therefore have bounded lifetime and bounded cumulative entropy throughput.

The empirical input is the per-bit conversion constant. Landauer furnishes the universal floor. The same calibrated model yields the finite replication entropy gap

$$\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k.$$

Keywords: molecular docking, exact resolution, structural rank, decision entropy, Landauer principle

## 1 Introduction

Exact molecular docking is treated as exact configuration resolution under constraints. The structural statements are finite counting, coordinate sufficiency, structural rank, decision-quotient entropy, and thermodynamic lower bounds. The abstract layer is a bounded decision system, represented in Lean by **Architecture** together with the canonical binary encoding **canonicalDP**. The molecular layer instantiates the same framework through holonomic constraint topology, cutoff-local interaction structure, and sampled docking families. The quotient object is closer to zero-error and confusability-based information than to average-case coding [23, 11, 13].

Landauer calibration supplies the universal floor [12, 1]. The central claim is that correctness, complexity, and thermodynamic cost are controlled by the same exact-resolution object: the decision quotient and its structural rank. For constrained molecular systems with  $N$  atoms and  $k$  independent holonomic constraints, the transported degree-of-freedom count is  $3N - k$ , and the exact-resolution floor scales with that remaining unconstrained dimension.

### 1.1 Central Result

The convergence theorem proved in Section 6 is:

$$\underbrace{\text{DOF}(A) = 1}_{\text{one-coordinate regime}} \iff \underbrace{\text{sr}\text{rank}(\text{canonicalDP}(A)) = 1}_{\text{structural rank}} \\ \iff \underbrace{\text{tractable sufficiency for canonicalDP}(A)}_{\text{complexity}} \iff \underbrace{\text{minimum per-cycle thermodynamic cost}}_{\text{physics}}$$

An imported coherence theorem gives a separate single-source reading of the same rank-1 point:

$$\text{SSOT}(A) \iff \text{DOF}(A) = 1,$$

where  $\text{SSOT}(A)$  denotes the coherent single-source condition that one locus is authoritative, every remaining encoding is a derived view, and all reachable states remain coherent.

The structural-rank and thermodynamic clauses are central. The imported coherence statement is a companion interpretation of the same rank-1 regime.

The same chain has a direct molecular reading. Exact molecular resolution requires a sufficient coordinate set. Cutoff locality bounds the number of decision-relevant protein coordinates by the active site together with the ligand coordinates, and sampled docking preserves exact winners under explicit half-gap control. The resulting structural-rank bounds and thermodynamic floors are therefore not detached corollaries but the concrete molecular instantiation of the abstract theory.

## 1.2 Structural and Empirical Inputs

Theorem 2.4 fixes the finite-event statement of the model. Proposition 2.5 and Theorem 2.6 fix the traversal-rate statement for bounded regions. Theorems 2.7 and 2.8 fix the acquisition-event interface used by the canonical encoding. Landauer calibration enters only in Section 5, where the per-bit lower bound is converted into energy.

## 1.3 Contributions

1. **Exact Docking Semantics:** exact molecular docking is exact configuration resolution under constraints through sufficient coordinate sets, the decision quotient, and the canonical exact-resolution encoding.
2. **Complexity Boundary:** general exact sufficiency certification contains a hardness core, while cutoff locality bounds docking structural rank by active-site and ligand coordinates, and sampled docking admits theorem-backed exact/coarse winner preservation and inside-cutoff sufficiency under explicit gap and compatibility hypotheses.
3. **Thermodynamic Floor:** the canonical exact-resolution problem assigns one binary acquisition channel to each degree of freedom, the resulting structural rank equals that channel count, the decision entropy is bounded by the same count, and under Landauer calibration exact-resolution cost is bounded below by both  $\text{DOF}(A) k_B T \ln 2$  and  $k_B T H_{\text{nats}}(D)$ .
4. **Molecular Constraint Transport:** constrained molecular systems with  $N$  atoms and  $k$  independent holonomic constraints transport directly into the framework with effective dimension  $3N - k$ , binary constraint-status interface, and a Landauer-linear floor scaling with the remaining unconstrained coordinates.
5. **Nonideal Exact Resolution:** theorem-level mismatch and residual witnesses force effective per-bit floors strictly above Landauer, explicit binary mismatch and two-state residual witnesses each yield an additive one-unit overhead above the Landauer floor, and finite-capacity substrates therefore have bounded lifetime and bounded entropy throughput.
6. **Convergence and Universal Consequences:** the rank-1 regime is simultaneously the one-coordinate regime, the tractable sufficiency regime, the coherent single-source regime, and the thermodynamic ground state; the same model yields the finite replication entropy gap and the finite-budget no-collapse theorem.
7. **Finite Physical Acquisition:** bounded systems admit finitely many acquisition events, bounded regions admit finite acquisition rates, acquisition is represented by discrete transition events, and exact resolution requires a sufficient coordinate set.
8. **Concrete Substrate Instantiation:** thresholded one-bit readouts and two-level atomic transition systems instantiate the canonical binary interface used by the theorem chain.

**Physical significance.** Once finite acquisition is fixed as the physical interface, the degree-of-freedom count becomes simultaneously an interaction dimension, an entropy bound, and a Landauer-calibrated cost coordinate. The result concerns exact resolution structure in matter.

Informally: exact resolution must be paid for.

## 1.4 Scope

The mathematical structure links structural rank, quotient entropy, complexity, and thermodynamic cost. Theorems are stated abstractly for bounded decision systems represented by the Lean object `Architecture` and are instantiated concretely for constrained molecular systems, cutoff-local docking problems, and sampled docking families.

## 1.5 Organization

Section 2 defines the structural model, the finite-acquisition interface, and the exact docking semantics. Section 3 derives the quotient, structural-rank, and compression consequences of exact resolution. Section 4 states the hardness core, the cutoff-local low-rank regime, and the sampled docking preservation theorems. Section 5 derives the thermodynamic cost consequences. Section 6 states the convergence theorem and the remaining universal consequences. Section 7 situates the results relative to thermodynamics, information theory, and molecular computation. Appendix A records proof provenance.

# 2 Foundations of Exact Molecular Docking

The formal objects that carry the mathematical-physics content are a positive degree-of-freedom count, a canonical binary decision encoding, structural rank, and decision entropy.

## 2.1 Formal Object

**Definition 2.1** (Bounded Decision System). A *bounded decision system* is a finite bounded physical system equipped with a positive integer  $\text{DOF}(A)$ . The corresponding Lean object is named `Architecture`. The results used below depend only on the degree-of-freedom count and on the canonical decision encoding.

**Interpretation.**  $\text{DOF}(A)$  counts independent coordinates that can vary separately. Subsequent sections study what that count forces once one asks for exact resolution.

**Molecular instantiation.** In the docking setting, the bounded decision system is a constrained molecular configuration space together with the binding decision problem induced by the chosen interaction model. The abstract degree-of-freedom count is later instantiated by holonomic constraint topology and local interaction structure.

## 2.2 Degrees of Freedom

**Definition 2.2** (Degrees of Freedom). The quantity  $\text{DOF}(A) \in \mathbb{N}$  counts independent coordinates of variation in a bounded decision system  $A$ . In the mechanized development it is the local structural parameter attached to `Architecture`; later sections identify it exactly with the structural rank of a canonical decision problem.

**Operational meaning.** If  $\text{DOF}(A) = n$ , the system has  $n$  independent coordinates that must be resolved in the worst case by any exact resolver.

**Molecular instantiation.** For a constrained molecular system with  $N$  atoms and  $k$  independent holonomic constraints, the transported degree-of-freedom count is  $3N - k$ . Later sections combine this finite topological count with cutoff-local docking structure.

Proposition 2.3 (DOF Additivity). *For disjoint bounded decision systems  $A_1$  and  $A_2$ :*

$$\text{DOF}(A_1 \oplus A_2) = \text{DOF}(A_1) + \text{DOF}(A_2)$$

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*Proof.* Independent coordinate sets combine by disjoint union, so the coordinate count is additive under composition. ■

## 2.3 Finite Physical Acquisition

Theorem 2.4 (Counting Gap). *Let  $\varepsilon, C \in \mathbb{N}$  with  $\varepsilon > 0$  and  $C > 0$ . If each information-acquisition event consumes  $\varepsilon$  discrete cost units, then*

$$\varepsilon \cdot N \leq C \implies N \leq C.$$

*Equivalently, any bounded system with positive per-event cost admits only finitely many acquisition events.*

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*Proof.* In  $\mathbb{N}$  every positive integer is at least one, so  $N = 1 \cdot N \leq \varepsilon N \leq C$ . ■

Theorem 2.4 fixes the finite-event statement of the model. Proposition 2.5 and Theorem 2.6 fix the geometric acquisition bound. Theorems 2.7 and 2.8 fix the acquisition-event interface. Landauer calibration is applied only after that interface is fixed.

Proposition 2.5 (Bounded Region). *A bounded physical region is characterized by diameter  $d > 0$  and signal speed  $c > 0$ . Its maximum information-acquisition rate is  $c/d$  events per unit time.*

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Theorem 2.6 (Bounded Acquisition Rate). *For a bounded region with diameter  $d$ , signal speed  $c$ , and operating time  $T$ ,*

$$\text{acquisitions}(T) \leq \frac{cT}{d}.$$

*In particular, acquisition count is finite on finite horizons.*

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*Proof.* Signals require at least  $d/c$  time to traverse the region, so no more than  $c/d$  acquisition events can occur per unit time. ■

Theorem 2.7 (Discrete Acquisition). *In the imported bounded-acquisition model, information acquisition is counted by transition points of a finite discrete system. Acquisition counts are therefore discrete event counts.*

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*Proof.* The imported model represents a bounded physical decision process by a finite `DiscreteSystem`. Its acquisition count is `bitOperations`, which counts transition points along a run. ■

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<sup>1</sup> Lean: [L17](#), [L19](#)   <sup>2</sup> : [BA10](#)   <sup>3</sup> : [BA1](#)   <sup>4</sup> : [BA2](#)   <sup>5</sup> : [BA3](#)

Theorem 2.8 (One Transition, One Bit). *In the imported discrete acquisition model, each transition point contributes one unit to the bit-operation count. The canonical binary encoding therefore uses one Boolean coordinate per elementary acquisition event.*

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*Proof.* The imported theorem states that a transition point at time  $t$  contributes at least one unit to the acquisition count up to time  $t + 1$ . The model therefore treats each elementary transition as one Boolean acquisition event. ■

Together, Theorems 2.6, 2.7, and 2.8 identify the later binary decision encoding as the natural finite acquisition model. A bounded resolver acquires information through finitely many discrete events, and each such event contributes one elementary boolean distinction.

Theorem 2.9 (Resolution Requires a Sufficient Coordinate Set). *Any exact physical resolver for a decision problem must read a sufficient coordinate set. If fewer coordinates are read, there exist states indistinguishable to the resolver but requiring different optimal actions.*

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*Proof.* If the accessed coordinates are not sufficient, two states agree on every read coordinate while disagreeing on the optimal action. Any resolver limited to those coordinates must therefore err on at least one of the two states. ■

## 2.4 Canonical Decision Encoding

Definition 2.10 (Canonical Decision Problem). For a bounded decision system  $A$  with  $\text{DOF}(A) = n$ , the canonical decision problem

$$\text{canonicalDP}(A)$$

has state space  $\text{Fin } n \rightarrow \text{Bool}$  and action space  $\text{Fin } n \oplus \text{Unit}$ . Action  $\text{inl}(i)$  queries coordinate  $i$ ; the fallback action  $\text{inr}(\star)$  receives constant utility 1. Query action  $i$  receives utility 2 exactly when coordinate  $i$  is true and 0 otherwise.

The encoding is the exact Lean object `canonicalDP` in `Leverage/BridgeToDQ.lean`. The development uses this encoding as the exact finite-resolution object attached to the declared degree-of-freedom count. It assigns one binary acquisition channel to each degree of freedom and one query action to each channel. The next section identifies the structural rank of this object with that coordinate count.

**Docking reading.** The canonical exact-resolution problem records the distinctions that any exact docking resolver must preserve. The quotient of this problem is therefore the exact abstraction of docking correctness, not an auxiliary coding artifact.

## 2.5 Structural Rank

Definition 2.11 (Structural Rank). The *structural rank* of a finite decision problem is the cardinality of its relevant coordinate set, equivalently the size of any minimal sufficient set. It is the minimum interaction dimension that must be read to determine the optimal action exactly.

For the canonical decision problem attached to  $A$ , the relevant coordinate set is all of  $\text{Fin } \text{DOF}(A)$ , so the structural-rank problem is exactly matched to the local degree-of-freedom count.

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<sup>6</sup> : BA4    <sup>7</sup> : BA5

## 2.6 Decision Quotient and Entropy

Definition 2.12 (Decision Quotient). For a decision problem  $D$ , states are identified when they induce the same optimal-action set. The quotient space of these equivalence classes is the *decision quotient* of  $D$ .

Proposition 2.13 (Optimizer Quotient as Canonical Exact Abstraction). *Let  $\text{Opt} : S \rightarrow \mathcal{P}(A)$  be the optimizer map of a finite decision problem. In **Set**, the decision quotient is the coimage of  $\text{Opt}$ , canonically equivalent to  $\text{im}(\text{Opt})$  [14]. Any surjective decision-preserving summary factors through this quotient.*

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The quotient is the coarsest exact abstraction of the decision problem: it forgets only decision-irrelevant distinctions and preserves every distinction needed for exact action selection. The entropy and thermodynamic bounds below are stated for this canonical exact abstraction.

Definition 2.14 (Decision Entropy). Let  $\text{numOptClasses}(D)$  be the number of equivalence classes in the decision quotient. Two entropy normalizations are used:

$$H_{\text{bits}}(D) = \log_2 \text{numOptClasses}(D), \quad H_{\text{nats}}(D) = \log \text{numOptClasses}(D).$$

The physics results are naturally stated in nats because Landauer calibration contributes the factor  $k_B T$  per nat of resolved decision information [12, 1].

## 2.7 Formalization in Lean

The local degree-of-freedom object lives in **Leverage/Foundations.lean**, while the canonical decision encoding and its rank-identification theorems live in **Leverage/BridgeToDQ.lean**. Structural rank and decision entropy are formalized in the decision-quotient development.

# 3 Exact Resolution, Quotient Structure, and Compression

The theorems of this section are statements about the canonical exact-resolution encoding defined in Section 2. The degree-of-freedom count is exactly the interaction dimension of that encoded decision problem, and exact physical resolution must pay for that interaction dimension in discrete bit events. Structural rank is the count of irreducible coordinate reads required by exact resolution in that encoding.

## 3.1 Degree of Freedom Equals Structural Rank

Theorem 3.1 (DOF–Structural-Rank Identity). *For every bounded decision system  $A$ ,*

$$\text{srnk}(\text{canonicalDP}(A)) = \text{DOF}(A).$$

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*Proof.* Write  $n = \text{DOF}(A)$ . By Definition 2.10, the state space is  $\text{Fin } n \rightarrow \text{Bool}$ , query action  $\text{inl}(i)$  has utility 2 exactly when coordinate  $i$  is true and utility 0 otherwise, and the fallback action has utility 1. Fix any coordinate  $i$  and choose two states that agree everywhere except at  $i$ , with one state setting  $i$  to true and the other setting  $i$  to false; then  $\text{inl}(i)$  is optimal in the first state and not optimal in the second, so erasing coordinate  $i$  changes the optimizer. Thus every coordinate in  $\text{Fin } n$  is relevant, the relevant-coordinate set has cardinality  $n$ , and the structural rank is  $n$ . Substituting  $n = \text{DOF}(A)$  gives the claim. ■

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<sup>8</sup> : QT1-3, QT5, QT7    <sup>9</sup> : L43

Theorem 3.1 is an exact identity for the canonical encoding attached to the degree-of-freedom count.

Corollary 3.2 (Rank-One Regime). *For every bounded decision system  $A$ ,*

$$\text{DOF}(A) = 1 \iff \text{srnk}(\text{canonicalDP}(A)) = 1.$$

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Corollary 3.3 (Higher-Rank Regime). *For every bounded decision system  $A$ ,*

$$\text{DOF}(A) > 1 \implies \text{srnk}(\text{canonicalDP}(A)) > 1.$$

11

Theorem 3.4 (Minimum Physical Bit Operations). *Any exact resolver for  $\text{canonicalDP}(A)$  requires at least  $\text{DOF}(A)$  elementary bit-acquisition events.*

12

*Proof.* By Theorem 2.9, exact resolution requires reading a sufficient coordinate set. The structural-rank theorem implies every sufficient set has cardinality at least  $\text{srnk}(\text{canonicalDP}(A))$ . Theorem 3.1 identifies that rank with  $\text{DOF}(A)$ , so at least  $\text{DOF}(A)$  coordinate reads are required. By Theorem 2.8, each read contributes one elementary bit-acquisition event. ■

### 3.2 Decision-Quotient Size

Theorem 3.5 (Decision-Class Bound). *For the canonical binary decision problem attached to a bounded decision system  $A$ ,*

$$\text{numOptClasses}(\text{canonicalDP}(A)) \leq 2^{\text{DOF}(A)}.$$

13

*Proof.* For binary coordinate spaces, the number of distinct optimal-action classes is at most  $2^{\text{srnk}}$ . Apply that theorem to the canonical encoding and substitute 3.1. ■

Theorem 3.6 (Decision-Entropy Bound). *For the canonical binary decision problem attached to a bounded decision system  $A$ ,*

$$H_{\text{bits}}(\text{canonicalDP}(A)) \leq \text{DOF}(A), \quad H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{DOF}(A) \ln 2.$$

14

*Proof.* The bit-entropy statement is the entropy-rank inequality for binary coordinate spaces, again composed with Theorem 3.1. The nat-entropy statement is obtained by multiplying by  $\ln 2$ . ■

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<sup>10</sup> : L51   <sup>11</sup> : L46   <sup>12</sup> : BA5-6, L43   <sup>13</sup> : IT4, L43   <sup>14</sup> : IT3, L43



### 3.3 Finite Compression Bridge

The next proposition packages the finite bridge in direct compression language: a finite Hamiltonian induces a deterministic tie-broken compression relation, and the paper1 fiber moment becomes the exact collision moment of that relation.

Proposition 3.7 (Finite Compression-Relation Bridge).<sup>15</sup> *Let  $H(c, \bar{c})$  be a finite compression Hamiltonian. Write*

$$R_H^{\min}(c, \bar{c}) \iff \bar{c} \in \arg \min_{\bar{c}'} H(c, \bar{c}'), \quad R_H^{\text{tb}}(c, \bar{c}) \iff \bar{c} \text{ is the least minimizer.}$$

For

$$M_H(\bar{c}) := \left| \left\{ c : R_H^{\text{tb}}(c, \bar{c}) \right\} \right|,$$

and every  $s \in \mathbb{N}$ ,

$$\left| \left\{ (\bar{c}, (c_i)_{i < s}) : \forall i, R_H^{\text{tb}}(c_i, \bar{c}) \right\} \right| = \sum_{\bar{c}} M_H(\bar{c})^s.$$

If moreover

$$\left| \left\{ c : R_H^{\min}(c, \bar{c}) \right\} \right| \leq 2^b \quad \text{for every } \bar{c},$$

then the induced tie-broken encoder has zero identity debt at budget  $b$ .

*Proof.* The first identity is the exact finite shared-codeword/fiber-moment theorem for the least-minimizer relation induced by  $H$ . The second statement uses that each tie-broken fiber sits inside the corresponding raw argmin fiber, so a uniform raw argmin bound transfers to the deterministic tie-broken encoder. ■

The compression bridge is the point of contact with the Landauer chain: in the exact-resolution reading of the canonical model, thermodynamic cost is the combinatorial cost of avoiding encoder collisions, because zero identity debt reduces to a uniform argmin-fiber bound, and that finite fiber-size condition is exactly what the energy–information theorem charges.

In docking language, the same finite fiber structure records when distinct molecular configurations remain exactly distinguishable under the binding decision relation.

Informally: to avoid collisions is to pay for distinctions.

### 3.4 Formalization

The structural-rank bridge is formalized in `Leverage/BridgeToDQ.lean`; the finite compression bridge is formalized in `Leverage/ColumnComplexityBridge.lean`; and the minimum-bit and entropy bounds are formalized in the decision-quotient physics and information development. These are the objects used directly by the thermodynamic theorems of the next section.

## 4 Complexity Boundary of Exact Molecular Docking

Exact molecular docking has a genuine tractability boundary. General exact sufficiency certification already contains a hardness core, while molecular locality supplies a low-rank regime in which exact resolution is structurally controlled. The claims in this section isolate that boundary before the thermodynamic lower bounds are applied.

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<sup>15</sup> : L57–58

## 4.1 General Hardness Core

Theorem 4.1 (General Hardness Core for Exact Sufficiency).<sup>16</sup> *For every Boolean formula  $\phi$ , the empty coordinate set is sufficient for the reduction problem induced by  $\phi$  if and only if  $\phi$  is a tautology.*

*Proof.* The reduction theorem is exact: tautology is encoded as sufficiency of the empty coordinate set for the induced decision problem. The equivalence therefore supplies the formal hardness core for exact sufficiency certification. ■

Theorem 4.2 (Maximal Structural Rank in the Hard Family).<sup>17</sup> *Let  $n > 0$ , and let  $\phi$  be a non-tautology over  $n$  Boolean variables. Then the many-coordinate reduction family has structural rank exactly  $n$ .*

*Proof.* Every coordinate is relevant in the non-tautology branch of the strengthened reduction family. Structural rank is the cardinality of the relevant-coordinate set, so the rank is exactly  $n$ . ■

The hard family therefore witnesses full interaction dimensionality: exact sufficiency can force the decision boundary to depend on every available coordinate.

## 4.2 Cutoff-Local Docking Regime

Theorem 4.3 (Cutoff-Local Structural-Rank Bound for Exact Docking).<sup>18</sup> *Let  $P_{\text{rel}}$  be the number of protein atoms within the cutoff radius of the binding site, and let  $L$  be the number of ligand atoms. Under the strict-optimum and outside-cutoff boundedness hypotheses,*

$$\text{srnk}(D_{\text{dock}}) \leq 3P_{\text{rel}} + 3L.$$

*Proof.* Any protein coordinate that remains decision-relevant must come from an atom within the cutoff radius, while every ligand coordinate remains available. Structural rank is therefore bounded by three coordinates for each cutoff-local protein atom together with three coordinates for each ligand atom. ■

Corollary 4.4 (Bounded-Pocket Low-Rank Regime).<sup>19</sup> *If at most  $K$  protein atoms lie within the cutoff radius of the binding site and the ligand has at most  $L$  atoms, then*

$$\text{srnk}(D_{\text{dock}}) \leq 3K + 3L.$$

*Proof.* Substitute the pocket-size and ligand-size bounds into Theorem 4.3. ■

Informally: bounded pockets bound exact difficulty.

## 4.3 Sampled Docking

Theorem 4.5 (Sampled Exact-Coarse Winner Preservation).<sup>20</sup> *For a finite sampled docking problem, let  $a_*$  be a strict exact winner at a sampled state  $s$ . If the coarse score differs from the exact score by at most  $\delta$  on every sampled action at  $s$ , and*

$$\delta < \frac{1}{2} \text{StrictUtilityGap}(a_*, s),$$

*then the exact and coarse optimal-action sets agree at  $s$ .*

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<sup>16</sup> : L63   <sup>17</sup> : L64   <sup>18</sup> : L66   <sup>19</sup> : L65   <sup>20</sup> : DQ2

*Proof.* The theorem is the sampled docking specialization of the strict half-gap invariance principle: a perturbation smaller than half the exact winner’s strict margin cannot change the optimal set. ■

Theorem 4.6 (Inside-Cutoff Sufficiency for Sampled Docking).<sup>21</sup> *Under the cutoff boundedness, sampled-optimum capture, coordinate-compatibility, and injectivity hypotheses, the retained coordinate set consisting of inside-cutoff protein coordinates together with all ligand coordinates is sufficient for the sampled restricted docking problem.*

*Proof.* Cutoff locality forces every relevant sampled coordinate into the retained set. The compatibility and injectivity hypotheses then lift the retained-set relevance bound into a sufficiency theorem for the sampled restricted problem. ■

## 4.4 Formalization

The local paper3 docking bridge theorems live in `Leverage/DockingTheoryBridge.lean`. They expose the general exact-sufficiency hardness core, the maximal-rank hard family, the cutoff-local structural-rank bounds for molecular docking, and the sampled exact/coarse preservation and sufficiency theorems used in the molecular development.

# 5 Thermodynamic Cost of Exact Molecular Docking

The preceding sections identified the quotient and structural-rank boundary underlying exact molecular docking. The thermodynamic theorems of this section compose three inputs: minimum bit operations, the entropy-rank bound, and a positive per-bit lower bound. The abstract statements hold for bounded decision systems, and the constrained-molecular corollaries transport them to holonomic topologies and binding-resolution problems. Landauer furnishes the universal floor for the conversion constant.

## 5.1 Landauer-Linear Resolution Cost

Theorem 5.1 (Rank Controls Exact-Resolution Cost).<sup>22</sup> *Let  $A$  be a bounded decision system and let  $M$  be a thermodynamic model with positive per-bit conversion constant. Then*

$$M.\text{joulesPerBit} \cdot \text{DOF}(A) \leq \text{energyLowerBound}(M, \text{DOF}(A)).$$

*In particular, exact-resolution cost is at least linear in the degree-of-freedom count.*

*Proof.* Theorem 3.4 gives a lower bound of  $\text{DOF}(A)$  elementary bit-acquisition events for exact resolution. The bounded-acquisition energy theorem then converts that bit lower bound into the displayed energy lower bound. ■

Theorem 5.2 (Rank-One Ground State).<sup>23</sup> *If  $\text{DOF}(A) = 1$ , then every exact-resolution cycle for the canonical problem has energy at least one Landauer unit. If  $\text{DOF}(A) > 1$ , then the system lies strictly above that ground state.*

*Proof.* The rank-one statement is exactly the imported ground-state theorem for structural rank 1 (BA8). For the higher-rank regime, Corollary 3.3 gives  $\text{srnk} > 1$ , and Theorem 5.1 then places the resulting exact-resolution cycle strictly above the one-Landauer-unit floor. ■

Informally: rank one is the ground state.

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<sup>21</sup> : DQ1    <sup>22</sup> : BA6-7, L43    <sup>23</sup> : BA8, L54-55

## 5.2 Energy–Information Duality

Theorem 5.3 (Energy–Information Duality).<sup>24</sup> *Let  $D = \text{canonicalDP}(A)$ , and let  $E$  be the realized energy for one exact-resolution cycle. If Landauer calibration holds at positive Boltzmann constant and temperature, then*

$$E \geq k_B T H_{\text{nats}}(D).$$

*Equivalently, the minimum exact-resolution cost is at least  $k_B T$  times the natural-log entropy of the decision quotient.*

*Proof.* The entropy-rank inequality gives

$$H_{\text{nats}}(D) \leq \text{DOF}(A) \ln 2$$

by Theorem 3.6. Theorem 5.1 gives the complementary lower bound

$$E \geq \text{DOF}(A) k_B T \ln 2.$$

Comparing the two right-hand sides yields the announced inequality. ■

Theorem 5.3 composes the entropy-rank inequality with the per-bit lower bound from Theorem 5.1.

Informally: the quotient fixing correctness also fixes cost.

Corollary 5.4 (Unique Minimum-Cost Regime).<sup>25</sup> *Among bounded decision systems in the canonical binary encoding, the unique minimum-cost regime is  $\text{DOF}(A) = 1$ .*

*Proof.* Theorem 5.2 identifies  $\text{DOF}(A) = 1$  as the one-Landauer-unit ground state, while every bounded decision system with more than one degree of freedom lies strictly above it. ■

## 5.3 Finite-Time and Budget Bounds

Theorem 5.5 (Exact-Resolution Time Lower Bound).<sup>26</sup> *Let  $A$  be a bounded decision system, and let  $I$  be a sufficient coordinate set for  $\text{canonicalDP}(A)$ . Suppose  $A$  is resolved inside a bounded region of diameter  $d$  and signal speed  $c$  over operating horizon  $\tau$ , and suppose*

$$|I| \leq \frac{c\tau}{d}.$$

*Then*

$$\text{DOF}(A) \leq \frac{c\tau}{d}.$$

*Proof.* Theorem 3.4 gives a lower bound of  $\text{DOF}(A)$  elementary acquisition events for exact resolution. Theorem 2.6 bounds the total number of acquisition events on horizon  $\tau$  by  $c\tau/d$ . Therefore exact resolution on that horizon requires  $\text{DOF}(A) \leq c\tau/d$ . ■

Theorem 5.6 (Decision-Class Bound from Spacetime and Energy Budget).<sup>27</sup> *Let  $D = \text{canonicalDP}(A)$ , and let  $I$  be a sufficient coordinate set for  $D$ . Suppose*

$$|I| \leq \frac{c\tau}{d}$$

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<sup>24</sup> : EI1, IT3, L43    <sup>25</sup> : BA8, L54–55    <sup>26</sup> : BA1–2, BA5–6, L43    <sup>27</sup> : BA1–2, BA5–6, IT3–4, EI1, L43

inside a bounded region of diameter  $d$  and signal speed  $c$  over operating horizon  $\tau$ , and let  $E$  satisfy

$$E \geq \text{DOF}(A) k_B \Theta \ln 2 \quad (\Theta > 0).$$

Then

$$\text{numOptClasses}(D) \leq 2^{c\tau/d} \quad \text{and} \quad \text{numOptClasses}(D) \leq \exp\left(\frac{E}{k_B \Theta}\right).$$

Consequently,

$$\text{numOptClasses}(D) \leq \min\left(2^{c\tau/d}, \exp\left(\frac{E}{k_B \Theta}\right)\right).$$

*Proof.* By Theorem 5.5, exact resolution on horizon  $\tau$  requires  $\text{DOF}(A) \leq c\tau/d$ . Theorem 3.5 gives

$$\text{numOptClasses}(D) \leq 2^{\text{DOF}(A)} \leq 2^{c\tau/d}.$$

Theorem 5.3 gives

$$E \geq k_B \Theta H_{\text{nats}}(D) = k_B \Theta \ln(\text{numOptClasses}(D)).$$

Hence

$$\ln(\text{numOptClasses}(D)) \leq \frac{E}{k_B \Theta},$$

which is equivalent to

$$\text{numOptClasses}(D) \leq \exp\left(\frac{E}{k_B \Theta}\right).$$

Taking the smaller of the two upper bounds gives the final statement. ■

Corollary 5.7 (Decision-Entropy Bound from Spacetime and Energy Budget).<sup>28</sup> Let  $D = \text{canonicalDP}(A)$ , and let  $I$  be a sufficient coordinate set for  $D$ . Suppose

$$|I| \leq \frac{c\tau}{d}$$

inside a bounded region of diameter  $d$  and signal speed  $c$  over operating horizon  $\tau$ , and let  $E$  satisfy

$$E \geq \text{DOF}(A) k_B \Theta \ln 2 \quad (\Theta > 0).$$

Then

$$H_{\text{bits}}(D) \leq \frac{c\tau}{d} \quad \text{and} \quad H_{\text{nats}}(D) \leq \min\left(\frac{c\tau}{d} \ln 2, \frac{E}{k_B \Theta}\right).$$

*Proof.* Theorem 5.5 gives  $\text{DOF}(A) \leq c\tau/d$ . Theorem 3.6 gives

$$H_{\text{bits}}(D) \leq \text{DOF}(A), \quad H_{\text{nats}}(D) \leq \text{DOF}(A) \ln 2.$$

Hence

$$H_{\text{bits}}(D) \leq \frac{c\tau}{d}, \quad H_{\text{nats}}(D) \leq \frac{c\tau}{d} \ln 2.$$

Theorem 5.3 also gives

$$H_{\text{nats}}(D) \leq \frac{E}{k_B \Theta}.$$

Taking the smaller of the two nat-valued upper bounds gives the final statement. ■

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<sup>28</sup> : BA1-2, BA5-6, IT3, EI1, L43

Corollary 5.8 (Independent Composition Budget Law).<sup>29</sup> *Let  $A_1$  and  $A_2$  be disjoint bounded decision systems. Suppose the composite system  $A_1 \oplus A_2$  is resolved inside a bounded region of diameter  $d$  and signal speed  $c$  over operating horizon  $\tau$ , and suppose some sufficient coordinate set for  $\text{canonicalDP}(A_1 \oplus A_2)$  has cardinality at most  $c\tau/d$ . Then*

$$\text{DOF}(A_1) + \text{DOF}(A_2) \leq \frac{c\tau}{d}$$

*and for any thermodynamic model with positive per-bit conversion constant,*

$$\text{joulesPerBit} \cdot (\text{DOF}(A_1) + \text{DOF}(A_2)) \leq \text{energyLowerBound}\left(M, \frac{c\tau}{d}\right).$$

*Proof.* Proposition 2.3 gives

$$\text{DOF}(A_1 \oplus A_2) = \text{DOF}(A_1) + \text{DOF}(A_2).$$

Apply Theorem 5.5 and Theorem 5.1 to the composite system and substitute the additive degree-of-freedom identity. ■

## 5.4 Worked Examples

Two toy canonical systems fix the scale of the bound.

**One coordinate in the canonical encoding.** Let  $A$  satisfy  $\text{DOF}(A) = 1$ , and write  $D = \text{canonicalDP}(A)$ . Then the state space of  $D$  has two states. The false state has optimal set  $\{\text{inr}(\star)\}$ , and the true state has optimal set  $\{\text{inl}(0)\}$ . The decision quotient therefore has two classes,  $H_{\text{nats}}(D) = \ln 2$ , and Theorem 5.3 gives

$$E \geq k_B T \ln 2.$$

This is the rank-1 ground regime.

**Two coordinates in the canonical encoding.** Let  $A$  satisfy  $\text{DOF}(A) = 2$ , and write  $D = \text{canonicalDP}(A)$ . Then the four states of  $D$  have optimal sets  $\{\text{inr}(\star)\}$ ,  $\{\text{inl}(0)\}$ ,  $\{\text{inl}(1)\}$ , and  $\{\text{inl}(0), \text{inl}(1)\}$ . The decision quotient therefore has four classes,  $H_{\text{nats}}(D) = \ln 4 = 2 \ln 2$ , and Theorem 5.3 gives

$$E \geq 2k_B T \ln 2.$$

Relative to the one-coordinate case, the minimum exact-resolution cost doubles.

## 5.5 Concrete Substrate Instantiation

Proposition 5.9 (Threshold Channel Realization).<sup>30</sup> *Fix a threshold  $\tau$  and a sampled substrate observable  $x_t \in \mathbb{R}$ . The induced readout bit*

$$b_t = \mathbf{1}[x_t \geq \tau]$$

*has binary state space  $\{0, 1\}$ . A readout flip  $b_{t+1} \neq b_t$  is equivalent to a positive one-bit lower bound, and under a positive per-bit conversion constant it implies a positive energy lower bound.*

<sup>29</sup> : L17, L19, L43, BA1-2, BA5-7    <sup>30</sup> : CV7-9

Proposition 5.10 (Two-Level Atomic Realization).<sup>31</sup> *Let  $c_0$  and  $c_1$  be atomic configurations with distinct orbital occupancies or distinct energies. Then  $c_0 \neq c_1$ . Upward transitions from  $c_0$  to  $c_1$  require positive energy input, and downward transitions release positive energy. A choice of labels  $c_0 \mapsto 0$  and  $c_1 \mapsto 1$  therefore gives a physical binary readout layer.*

Thresholded one-bit readouts and two-level atomic transitions instantiate the same binary interface [2, 19, 7, 21]. A  $k$ -channel substrate has joint readout state in  $\{0, 1\}^k$ , and the canonical state space  $\text{Fin } k \rightarrow \text{Bool}$  is the same object written in indexed form.

## 5.6 Substrate Time Law

Proposition 5.11 (Substrate Step is Unit Interface Time).<sup>32</sup> *For any substrate model whose observed interface obeys decision ticks, every one-step substrate evolution realizes a decision event and advances interface time by one unit. The tick law is independent of substrate tag.*

## 5.7 Strict Overhead Above Landauer

Proposition 5.12 (Theorem-Level Strict Overhead Branches).<sup>33</sup> *Let  $W$  be a decomposed process model. If the mismatch term is instantiated by a theorem-level distribution-mismatch witness, then the effective per-bit lower bound of  $W$  is strictly above the Landauer floor. If the residual term is instantiated by a theorem-level finite discrete residual witness, the same strict inequality holds. For any sufficient coordinate set  $I$  of  $\text{canonicalDP}(A)$ , either branch therefore yields an exact-resolution energy lower bound strictly above  $\text{DOF}(A) k_B T \ln 2$ .*

Proposition 5.13 (Finite Discrete Residual Witness).<sup>34</sup> *Let a finite computational-state process admit a positive forward edge together with decision-relevant asymmetry. Then the theorem-level discrete residual lower bound is positive. If this witness is used as the residual term of a decomposed process model, the effective per-bit lower bound is strictly above the Landauer floor.*

Proposition 5.14 (Explicit Two-State Residual Example).<sup>35</sup> *There exists a two-state irreversible residual witness with one positive forward edge and zero reverse edge. The induced residual lower-bound term is exactly one nat-valued overhead unit. Any decomposed process model that uses this witness as its residual term therefore satisfies*

$$\text{landauerJoulesPerBit}(k_B, T) + 1 \leq W.\text{effectiveModel}.\text{joulesPerBit},$$

and for any sufficient coordinate set  $I$  of  $\text{canonicalDP}(A)$ ,

$$\text{DOF}(A) (k_B T \ln 2 + 1) \leq \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

The same example yields the strengthened energy-information inequality

$$\frac{k_B T \ln 2 + 1}{\ln 2} H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

Proposition 5.15 (Repeated Residual-Example Work Law).<sup>36</sup> *For any  $m \in \mathbb{N}$ , repeated exact-resolution cycles under the same explicit two-state residual witness satisfy*

$$m \frac{k_B T \ln 2 + 1}{\ln 2} H_{\text{nats}}(\text{canonicalDP}(A)) \leq m \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

The required cumulative work therefore grows linearly with cycle count.

<sup>31</sup> : AC1, AC3-4    <sup>32</sup> : DT22-24    <sup>33</sup> : WM6, WR10, WP6, BA5-6, L43    <sup>34</sup> : WR6-7, WR10    <sup>35</sup> : WR11-12, IT3, BA5-6, L43  
<sup>36</sup> : WR11-12, IT3, BA5-6, L43

Proposition 5.16 (Unified Energy–Information Hierarchy). <sup>37</sup> *Let  $I$  be a sufficient coordinate set for  $\text{canonicalDP}(A)$ . Then the ideal Landauer-calibrated floor satisfies*

$$k_B T H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{energyLowerBound}(M, |I|)$$

*whenever the declared model  $M$  dominates the Landauer floor. Under the explicit binary mismatch example and the explicit two-state residual example, the strengthened coefficient*

$$\frac{k_B T \ln 2 + 1}{\ln 2}$$

*replaces  $k_B T$ :*

$$\frac{k_B T \ln 2 + 1}{\ln 2} H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{energyLowerBound}(W_{\text{mm}}.\text{effectiveModel}, |I|),$$

$$\frac{k_B T \ln 2 + 1}{\ln 2} H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{energyLowerBound}(W_{\text{res}}.\text{effectiveModel}, |I|).$$

Proposition 5.17 (Structural-Resource Overhead). <sup>38</sup> *Let  $W$  be a decomposed process model and let  $r$  be a declared structural resource. If  $r$  is lower-bounded by the mismatch term, then the effective per-bit lower bound dominates the Landauer floor plus  $r$ . For any sufficient coordinate set  $I$  of  $\text{canonicalDP}(A)$ ,*

$$\text{energyLowerBound}(W.\text{base}, |I|) + r \text{DOF}(A) \leq \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

Proposition 5.18 (Canonical Wolpert Grounding Bundle). <sup>39</sup> *Let  $I$  be a nonempty sufficient coordinate set for  $\text{canonicalDP}(A)$ , and let  $W$  be a decomposed process model whose base lower bound dominates the Landauer floor. Then*

$$\text{DOF}(A) \leq |I|,$$

$$W.\text{effectiveModel}.\text{joulesPerBit} \cdot \text{DOF}(A) \leq \text{energyLowerBound}(W.\text{effectiveModel}, |I|),$$

*and*

$$0 < \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

Proposition 5.19 (Strict Canonical Energy Above the Landauer Floor). <sup>40</sup> *Let  $I$  be a sufficient coordinate set for  $\text{canonicalDP}(A)$ , and let  $W$  be a decomposed process model whose base lower bound dominates the Landauer floor. If either theorem-level Wolpert branch applies to  $W$ , then*

$$\text{DOF}(A) k_B T \ln 2 < \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

Proposition 5.20 (Explicit Binary Mismatch Example). <sup>41</sup> *Fix the actual input distribution*

$$p(1) = \frac{3}{4}, \quad p(0) = \frac{1}{4},$$

*and the designed distribution*

$$q(1) = \frac{1}{4}, \quad q(0) = \frac{3}{4}.$$

*The induced mismatch lower-bound term is at least one nat-valued overhead unit. Any decomposed process model that uses this witness as its mismatch term therefore satisfies*

$$\text{landauerJoulesPerBit}(k_B, T) + 1 \leq W.\text{effectiveModel}.\text{joulesPerBit},$$

*and for any sufficient coordinate set  $I$  of  $\text{canonicalDP}(A)$ ,*

$$\text{DOF}(A) (k_B T \ln 2 + 1) \leq \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

<sup>37</sup> : IT3, WM4, WR11–12, WP2, BA5–6, L43    <sup>38</sup> : WP7–8, BA5–6, L43    <sup>39</sup> : WP9, BA5–6, L43    <sup>40</sup> : WP6, BA5–6, L43  
<sup>41</sup> : WM4, WP2, BA5–6, L43



Proposition 5.21 (Binary Mismatch Strengthens the Energy–Information Coefficient).<sup>42</sup> *Under the same explicit binary mismatch witness,*

$$\frac{k_B T \ln 2 + 1}{\ln 2} H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

*The coefficient is strictly larger than  $k_B T$ .*

Proposition 5.22 (Repeated Binary Mismatch Work Law).<sup>43</sup> *For any  $m \in \mathbb{N}$ , repeated exact-resolution cycles under the same explicit binary mismatch witness satisfy*

$$m \frac{k_B T \ln 2 + 1}{\ln 2} H_{\text{nats}}(\text{canonicalDP}(A)) \leq m \text{energyLowerBound}(W.\text{effectiveModel}, |I|).$$

*The required cumulative work therefore grows linearly with cycle count.*

## 5.8 Cyclewise Heat and Lifetime

Proposition 5.23 (Positive Heat and Bounded Lifetime).<sup>44</sup> *In the substrate heat-capacity model, every computational cycle generates positive heat, cumulative heat grows linearly with cycle count, heat above capacity causes degradation, and finite integrity together with finite heat capacity yields bounded lifetime.*

Proposition 5.24 (Finite Lifetime Throughput Bound).<sup>45</sup> *Let  $s$  be a finite substrate with lifetime ceiling  $\text{maxCycles}(s)$ . For any run of  $m$  exact-resolution cycles with*

$$m \leq \text{maxCycles}(s),$$

*the cumulative nat-valued decision entropy processed by  $\text{canonicalDP}(A)$  satisfies*

$$m H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{maxCycles}(s) \text{DOF}(A) \ln 2.$$

Proposition 5.25 (Speed-Heat Tradeoff).<sup>46</sup> *In the same substrate model, faster computation yields a larger heat rate. Once heat rate exceeds substrate capacity, faster computation yields faster degradation.*

## 5.9 Constrained Molecular Application

Corollary 5.26 (RATTLE Holonomic-Constraint Landauer Floor).<sup>47</sup> *Let  $X$  be a finite RATTLE holonomic topology with  $N$  atoms and  $k$  independent constraints, where each constraint check is recorded as a binary satisfied/violated status. Then the full constraint-status observation space has cardinality*

$$2^k.$$

*Let  $A_X$  be the transported bounded decision system with*

$$\text{DOF}(A_X) = 3N - k.$$

*Then the canonical exact-resolution problem satisfies*

$$\text{srnk}(\text{canonicalDP}(A_X)) = 3N - k.$$

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<sup>42</sup> : WM4, WP2, IT3, BA5-6, L43    <sup>43</sup> : WM4, WP2, IT3, BA5-6, L43    <sup>44</sup> : SE1-5    <sup>45</sup> : SE5, IT3, L43    <sup>46</sup> : SE6    <sup>47</sup> : L60-62

Moreover, for any sufficient coordinate set  $I$  for  $\text{canonicalDP}(A_X)$  and any thermodynamic model with positive per-bit conversion constant,

$$M.\text{joulesPerBit} \cdot (3N - k) \leq \text{energyLowerBound}(M, |I|).$$

In particular, the per-cycle exact-resolution floor scales linearly with the unconstrained molecular dimension.

*Proof.* The RATTLE holonomic status register is a  $k$ -bit binary interface by the finite cardinality theorem. The transported architecture has degree of freedom exactly  $3N - k$  by construction. The local bridge theorem identifies the structural rank of the canonical exact-resolution problem with that same count, and the local energy lower bound then gives the displayed Landauer-linear floor. ■

Informally: every independent coordinate raises the floor.

## 5.10 Optimal-Transport Witness

The Landauer route lower-bounds exact resolution through irreversible bit acquisition. A complementary witness measures separation between future distributions on the integrity space  $\{\text{intact}, \text{compromised}\}$ . It does not replace the main proof chain, but it supplies an independent transport-theoretic signal that multiple distinguishable futures have nonzero cost.

Remark 5.27 (Optimal-Transport Bridge).<sup>48</sup> The same separation admits an independent transport-cost witness on the two-state integrity space  $\{\text{intact}, \text{compromised}\}$ . The diagonal coupling has zero transport cost in the single-future regime (W1). Any coupling with off-diagonal mass has positive transport cost (W2). If the intact future mass dominates the compromised future mass, the intact state minimizes total transport to the future distribution (W3). If both future states carry positive mass, then transport from either pure state is strictly positive (W4). Multiple distinguishable futures therefore force positive transport cost independently of the Landauer route.

A transport witness and the Landauer witness emphasize different structures: one counts irreducible coordinate reads, while the other measures geometric separation of future mass. In the two-state integrity model they point in the same direction, since a single future has zero transport cost whereas genuinely split futures force strictly positive transport cost.

## 5.11 Interpretation

If degree of freedom is read as the number of independent physical coordinates that can vary separately, then lower DOF means lower exact-resolution cost because fewer independent coordinates must be resolved. The constrained-molecular corollary makes that transport explicit for the finite count  $3N - k$ .

## 5.12 Formalization

The local bridge from degree of freedom to structural rank is formalized in `Leverage/BridgeToDQ.lean`, including the direct finite RATTLE transport with effective dimension  $3N - k$ . The finite holonomic-constraint counting layer lives in `Computation/GeometricConstraints.lean`. The physical acquisition and Landauer theorems are imported from the decision-quotient physics stack, in particular `Physics/BoundedAcquisition.lean` and `ThermodynamicLift.lean`. The role of the local

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<sup>48</sup> : W1-4

Architecture object in this section is to provide the coordinate count transported into those theorems.

## 6 Convergence and Universal Consequences

Degree of freedom equals structural rank, and structural rank bounds decision entropy. The same rank-1 regime also appears as tractable sufficiency and minimum thermodynamic cost. The molecular instantiations above make the same point concrete for constrained molecular systems, while the remaining theorems in this section record the universal consequences that survive beyond the docking specialization.

### 6.1 Imported Coherence Reading

Theorem 6.1 (Coherent Single-Source Regime). <sup>49</sup> *A bounded decision system lies in the coherent unit-independent-rate regime if and only if  $\text{DOF}(A) = 1$ .*

Remark 6.2 (Imported Reading). In the imported coherence development, rank 1 means exactly one locus is authoritative, every remaining encoding is a derived view, and all reachable states remain coherent.

### 6.2 Structural Rank

Theorem 6.3 (Rank Identification). <sup>50</sup> *For every bounded decision system  $A$ ,*

$$\text{srnk}(\text{canonicalDP}(A)) = \text{DOF}(A).$$

### 6.3 Tractability Boundary

Theorem 6.4 (Tractable Sufficiency at Rank One). <sup>51</sup> *In the canonical decision problem family, structural rank 1 is the tractable sufficiency regime, while higher structural rank enters the hard regime.*

### 6.4 Thermodynamic Selection

Theorem 6.5 (Thermodynamic Selection). <sup>52</sup> *In the canonical decision encoding, every bounded decision system with  $\text{DOF}(A) > 1$  lies strictly above the rank-1 Landauer ground state in per-cycle resolution cost.*

Remark 6.6 (Scope Note). The theorem uses only the rank identity together with Landauer calibration. Stronger hardness consequences require additional imported hypotheses.

### 6.5 Convergence Theorem

Theorem 6.7 (Convergence). <sup>53</sup> *For every bounded decision system  $A$ , the following conditions are equivalent:*

$$\text{DOF}(A) = 1 \iff \text{srnk}(\text{canonicalDP}(A)) = 1 \iff \text{tractable sufficiency for canonicalDP}(A) \iff \text{minimum per-cycle resolution cost}$$

<sup>49</sup> : Imported coherence theorem (ORA1) <sup>50</sup> : L43, L46, L51 <sup>51</sup> : Imported tractability theorem (L56, L53, L47)

<sup>52</sup> : L49, L54-55, BA8 <sup>53</sup> : L43-44, L47, L55

*The same rank-1 regime is simultaneously the exact one-coordinate regime, the tractable exact-certification regime, and the thermodynamic ground state. The imported coherence theorem gives an additional single-source interpretation of the same point.*

**Proof.**

1. Theorem 6.3 identifies  $\text{DOF}(A) = 1$  with structural rank 1.
2. Theorem 6.4 identifies structural rank 1 with the tractable sufficiency regime for the canonical family.
3. Theorem 6.5 identifies rank 1 as the unique minimum-cost thermodynamic regime.

Transitivity of logical equivalence completes the proof.

Theorem 6.1 supplies an imported single-source interpretation of the same rank-1 regime.

## 6.6 Formalization

The local bridge theorems live in `Leverage/BridgeToDQ.lean`; the coherence theorem is imported from `Ssot`; and the tractability and Landauer-cost theorems are imported from the decision-quotient development. Appendix A records source provenance.

## 6.7 Finite Replication Entropy Gap

The theorem below uses the rank-indexed entropy cost  $\Delta S_{\min}(r) = rk_B \ln 2$  from the calibrated exact-resolution model.

Theorem 6.8 (Finite Replication Entropy Gap).<sup>54</sup> *Let  $\Delta S_{\min}(r) = r \cdot k_B \ln 2$  be the rank-indexed minimal entropy production under Landauer calibration. For the rank-1 ground regime and any replicated rank- $k$  regime:*

$$\Delta S_{\min}(1) + k_B \ln k \leq \Delta S_{\min}(k)$$

*equivalently,  $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$ .*

*Proof.* The gap is  $(k - 1) \cdot k_B \ln 2$ . Since  $k \leq 2^{k-1}$  (L52), taking logs gives  $\ln k \leq (k - 1) \ln 2$ , so the gap is  $\geq k_B \ln k$ . ■

**Model class.**  $\Delta S_{\min}$  is the rank-indexed Landauer entropy cost in the calibrated exact-resolution model. The theorem uses the finite inequality  $k \leq 2^{k-1}$ . England’s 2013 result [8] is a stochastic-thermodynamic path-space theorem with detailed balance and far-from-equilibrium dynamics. The common term is the multiplicity penalty  $k_B \ln k$ .

## 6.8 Finite-Budget No-Collapse

Theorem 6.9 (Finite-Budget No-Collapse).<sup>55</sup> *Let  $B : \mathbb{N} \rightarrow \mathbb{N}$  be a budget profile, let  $\text{ops} : \mathbb{N} \rightarrow \mathbb{N}$  be a required-operation profile, and let  $\text{bitCost} > 0$  be the per-bit physical cost. If*

1.  *$B$  is globally bounded,*
2.  *$\text{ops}$  has an exponential lower bound, and*
3. *collapse means: for every input size  $n$ , some feasible bit budget realizes at least  $\text{ops}(n)$  operations within budget  $B(n)$ ,*

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<sup>54</sup> : L45    <sup>55</sup> : PH26

then no such physical collapse profile exists.

*Proof.* This is exactly the bounded-budget physical no-collapse theorem in the physical-hardness layer. Exponential growth eventually exceeds every fixed finite budget, and positive per-bit cost lifts that growth into an energy contradiction. ■

Remark 6.10 (Interpretation). Finite budget, positive event cost, and exponential exact-certification demand are jointly incompatible with a physical collapse model. Any stronger complexity-collapse conclusion requires an additional bridge from the chosen complexity claim to such a collapse profile.

## 7 Related Work

### 7.1 Landauer, Non-Equilibrium Thermodynamics, and Selection

Landauer’s principle gives the standard calibration from logically irreversible discrimination to minimum heat production and energy cost [12, 1]. Stochastic thermodynamics extends that floor to trajectory-level entropy production, work identities, and fluctuation relations [22, 26, 27, 10, 3]. Finite-time erasure and mismatch corrections sharpen the same theme for controlled nonequilibrium protocols [6, 20, 15]. The theorem chain above isolates a different object: a finite exact-resolution lower bound indexed by the number of independent coordinates that must be resolved to preserve the optimizer.

Relative to the Seifert and Van den Broeck–Esposito framework, the present model gives a non-asymptotic lower bound in terms of structural rank and decision-quotient entropy. Stochastic-thermodynamic frameworks resolve time-dependent nonequilibrium refinements that are outside the current finite exact-resolution model.

England’s 2013 result is a stochastic-thermodynamic path-space theorem with detailed balance and far-from-equilibrium dynamics [8]. The corresponding replication theorem in the calibrated exact-resolution model is a finite Landauer-counting statement. The common term is the multiplicity penalty  $k_B \ln k$ .

### 7.2 Zero-Error, Functional, and Quotient Information

The information object is closer to zero-error and confusability-based information theory than to average-case source coding [23, 11, 13, 4]. The central quantity is not full state entropy but the entropy of the decision quotient: how many distinct optimal-action classes survive after irrelevant coordinates are erased.

Function-relative information in physics and origins-of-life work also conditions information on successful function or selection [24, 28]. The exact-resolution object is narrower: coordinate erasure is admissible precisely when optimal-action correspondence is preserved. The rank-1 regime is therefore the one-coordinate exact-resolution regime, the tractable sufficiency regime, and the minimum calibrated-cost regime.

### 7.3 Categorical Quotients and Exact Abstraction

Quotienting states by equality of  $\text{Opt}$  is the standard coimage construction for the decision quotient of the optimizer map  $\text{Opt} : S \rightarrow \mathcal{P}(A)$  in **Set**, canonically equivalent to its image [14]. The novelty is not the existence of that quotient, but the theorem chain tying it to coordinate sufficiency, structural rank, decision entropy, and thermodynamic cost in one proof object.

## 7.4 Formal Source Provenance

Appendix A records source provenance for the stated claims. A Lean 4 proof file accompanies the archived artifact [5, 25]. Related precedents include verified computability and semantics developments in Coq and Isabelle [9, 18, 17] and certificate-carrying proof artifacts [16].

# 8 Conclusion

## 8.1 Summary

The central result is a complete abstract-plus-molecular theory of exact resolution. Theorem 6.7 identifies the rank-1 regime of the canonical decision encoding as simultaneously the one-coordinate regime, the tractable sufficiency regime, and the thermodynamic ground state. The molecular sections instantiate the same framework for constrained molecular systems, cutoff-local docking structure, and sampled docking.

The theorem package separates a structural part from an empirical calibration. The structural part is the finite acquisition chain, the canonical exact-resolution encoding, the exact-sufficiency hardness core, the identity  $\text{DOF}(A) = \text{srank}(\text{canonicalDP}(A))$ , the cutoff-local docking rank bounds, and the decision-entropy bound. The empirical inputs are bounded signal speed, the discrete transition interface used for acquisition, cutoff-local approximation control, and a positive per-bit lower bound. Landauer furnishes the universal floor.

### Main consequences:

- The exact identification of degree of freedom with structural rank in the canonical decision encoding.
- The general hardness core for exact sufficiency certification and the maximal-rank hard family.
- The cutoff-local structural-rank bound for molecular docking, the bounded-pocket low-rank regime, sampled exact/coarse winner preservation under a half-gap hypothesis, and inside-cutoff sufficiency for sampled docking under the stated compatibility assumptions.
- The energy-information theorem  $E \geq k_B T H_{\text{nats}}(D)$  for exact-resolution cost.
- The bounded-acquisition inequalities  $\text{DOF}(A) \leq c\tau/d$ , the induced decision-class and decision-entropy bounds from spacetime and energy budget, and the linear budget law for independent composition.
- The theorem-level strict-overhead branches above the Landauer floor, the finite discrete residual witness, the canonical Wolpert grounding bundle, the unified ideal/mismatch/residual energy-information hierarchy, the induced strict canonical energy separation above the Landauer-linear floor, explicit binary mismatch and two-state residual examples with additive one-unit overhead, strengthened energy-information coefficients, cumulative work laws, and the substrate step time law.
- The bounded-lifetime consequences of positive cyclewise heat in finite-capacity substrates and the resulting finite entropy-throughput ceiling.
- The unconditional thermodynamic selection statement that every higher-rank regime lies above the rank-1 Landauer ground state.

- The finite-budget no-collapse theorem: bounded budget, positive per-bit cost, and exponential lower-bound growth cannot coexist with physical collapse.
- The finite replication entropy gap:  $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$  in the calibrated exact-resolution model.

## 8.2 Scope

**1. Canonical-encoding scope:** the main theorems are exact for the canonical binary decision encoding attached to the bounded decision system. Extending the same conclusions to more general physical encodings requires an explicit transport argument.

**2. Calibration choice:** Landauer calibration supplies the physical conversion constant. Stronger substrate-dependent lower bounds belong to a different modeling layer.

**3. Replication theorem:** the finite replication entropy gap is a theorem of the calibrated exact-resolution model. England’s 2013 theorem belongs to a stochastic-thermodynamic path-space model.

**4. Finite-budget model class:** the no-collapse theorem is a statement about globally bounded budget profiles with positive per-bit cost and exponential lower-bound growth. Different collapse claims require explicit bridges into that profile language.

## 8.3 Final Remarks

Exact molecular docking is therefore governed by one formal chain rather than by separate semantic, complexity, and thermodynamic stories. Abstractly, exact resolution is governed by sufficient coordinate sets, structural rank, decision quotient, and calibrated thermodynamic floor. Concretely, constrained molecular systems instantiate the same framework through holonomic constraint topology, cutoff-local interaction structure, sampled exact/coarse stability, and direct Landauer-linear cost bounds.

The abstract layer is not decorative generality, and the molecular layer is not an afterthought. The abstract theorems state what exact resolution costs for any bounded decision system. The molecular instantiation shows that the same theorems govern exact docking, constrained molecular computation, and repeated exact molecular resolution in matter. Corollary 5.26 gives the direct RATTLE finite derivation: the constraint-status interface is a  $k$ -bit binary register, the effective coordinate count is  $3N - k$ , and the canonical Landauer floor scales linearly with that remaining unconstrained dimension. Remark B.1 isolates the remaining scope boundary precisely: the finite transport is proved once independence is specified, while derivation of that independence hypothesis from a concrete geometric constraint family remains additional work.

Appendix A records proof provenance.

# A Proof Provenance

This appendix reports claim traceability directly from source and generated mapping artifacts.

## A.1 Claim Coverage Matrix

| Paper claim                                   | Lean handle         |
|-----------------------------------------------|---------------------|
| Corollary 4.4: Bounded-Pocket Low-Rank Regime | <a href="#">L65</a> |

| Paper claim                                                                      | Lean handle                                                                                                                                                                     |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corollary 5.7: Decision-Entropy Bound from Spacetime and Energy Budget           | <a href="#">IT3</a> , <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">EI1</a> , <a href="#">L43</a>                         |
| Corollary 5.8: Independent Composition Budget Law                                | <a href="#">L17</a> , <a href="#">L19</a> , <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA7</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">L43</a>   |
| Corollary 5.26: RATTLE Holonomic-Constraint Landauer Floor                       | <a href="#">L60</a> , <a href="#">L61</a> , <a href="#">L62</a>                                                                                                                 |
| Corollary 5.4: Unique Minimum-Cost Regime                                        | <a href="#">BA8</a> , <a href="#">L54</a> , <a href="#">L55</a>                                                                                                                 |
| Corollary 3.3: Higher-Rank Regime                                                | <a href="#">L46</a> , <a href="#">L51</a>                                                                                                                                       |
| Corollary 3.2: Rank-One Regime                                                   | <a href="#">L51</a>                                                                                                                                                             |
| Proposition 5.10: Two-Level Atomic Realization                                   | <a href="#">AC1</a> , <a href="#">AC3</a> , <a href="#">AC4</a>                                                                                                                 |
| Proposition 5.22: Repeated Binary Mismatch Work Law                              | <a href="#">IT3</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WP2</a> , <a href="#">WM4</a> , <a href="#">L43</a>                                               |
| Proposition 5.21: Binary Mismatch Strengthens the Energy–Information Coefficient | <a href="#">IT3</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WP2</a> , <a href="#">WM4</a> , <a href="#">L43</a>                                               |
| Proposition 5.20: Explicit Binary Mismatch Example                               | <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WP2</a> , <a href="#">WM4</a> , <a href="#">L43</a>                                                                     |
| Proposition 5.15: Repeated Residual-Example Work Law                             | <a href="#">IT3</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WR12</a> , <a href="#">WR11</a> , <a href="#">L43</a>                                             |
| Proposition 5.14: Explicit Two-State Residual Example                            | <a href="#">IT3</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WR12</a> , <a href="#">WR11</a> , <a href="#">L43</a>                                             |
| Proposition 2.5: Bounded Region                                                  | <a href="#">BA1</a>                                                                                                                                                             |
| Proposition 5.18: Canonical Wolpert Grounding Bundle                             | <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WP9</a> , <a href="#">L43</a>                                                                                           |
| Definition 2.3: Bounded Decision System                                          | <a href="#">L17</a> , <a href="#">L19</a>                                                                                                                                       |
| Proposition 5.16: Unified Energy–Information Hierarchy                           | <a href="#">IT3</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WR12</a> , <a href="#">WP2</a> , <a href="#">WM4</a> , <a href="#">WR11</a> , <a href="#">L43</a> |
| Proposition 3.7: Finite Compression-Relation Bridge                              | <a href="#">L57</a> , <a href="#">L58</a>                                                                                                                                       |
| Proposition 5.13: Finite Discrete Residual Witness                               | <a href="#">WR10</a> , <a href="#">WR7</a> , <a href="#">WR6</a>                                                                                                                |
| Proposition 5.23: Positive Heat and Bounded Lifetime                             | <a href="#">SE1</a> , <a href="#">SE2</a> , <a href="#">SE3</a> , <a href="#">SE4</a> , <a href="#">SE5</a>                                                                     |
| Proposition 5.24: Finite Lifetime Throughput Bound                               | <a href="#">SE5</a> , <a href="#">IT3</a> , <a href="#">L43</a>                                                                                                                 |
| Definition 2.13: Canonical Decision Problem                                      | <a href="#">QT2</a> , <a href="#">QT7</a> , <a href="#">QT1</a> , <a href="#">QT3</a>                                                                                           |
| Proposition 5.25: Speed-Heat Tradeoff                                            | <a href="#">SE6</a>                                                                                                                                                             |
| Proposition 5.19: Strict Canonical Energy Above the Landauer Floor               | <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WP6</a> , <a href="#">L43</a>                                                                                           |
| Proposition 5.12: Theorem-Level Strict Overhead Branches                         | <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WM6</a> , <a href="#">WP6</a> , <a href="#">WR10</a> , <a href="#">L43</a>                                              |
| Proposition 5.17: Structural-Resource Overhead                                   | <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">WP8</a> , <a href="#">WP7</a> , <a href="#">L43</a>                                                                     |
| Proposition 5.11: Substrate Step is Unit Interface Time                          | <a href="#">DT23</a> , <a href="#">DT22</a> , <a href="#">DT24</a>                                                                                                              |
| Proposition 5.9: Threshold Channel Realization                                   | <a href="#">CV8</a> , <a href="#">CV9</a> , <a href="#">CV7</a>                                                                                                                 |
| Theorem 2.6: Bounded Acquisition Rate                                            | <a href="#">BA1</a> , <a href="#">BA2</a>                                                                                                                                       |
| Theorem 5.6: Decision-Class Bound from Spacetime and Energy Budget               | <a href="#">IT4</a> , <a href="#">IT3</a> , <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">EI1</a> , <a href="#">L43</a>   |
| Theorem 6.1: Coherent Single-Source Regime                                       | <a href="#">ORA1</a>                                                                                                                                                            |
| Theorem 2.4: Counting Gap                                                        | <a href="#">BA10</a>                                                                                                                                                            |
| Theorem 2.7: Discrete Acquisition                                                | <a href="#">BA3</a>                                                                                                                                                             |
| Theorem 3.1: DOF–Structural-Rank Identity                                        | <a href="#">L43</a>                                                                                                                                                             |
| Theorem 5.3: Energy–Information Duality                                          | <a href="#">IT3</a> , <a href="#">EI1</a> , <a href="#">L43</a>                                                                                                                 |
| Theorem 5.1: Rank Controls Exact-Resolution Cost                                 | <a href="#">BA7</a> , <a href="#">BA6</a> , <a href="#">L43</a>                                                                                                                 |



| Paper claim                                                       | Lean handle                                                                                                 |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Theorem 6.8: Finite Replication Entropy Gap                       | <a href="#">L45</a>                                                                                         |
| Theorem 3.6: Decision-Entropy Bound                               | <a href="#">IT3</a> , <a href="#">L43</a>                                                                   |
| Theorem 4.1: General Hardness Core for Exact Sufficiency          | <a href="#">L63</a>                                                                                         |
| Theorem 6.9: Finite-Budget No-Collapse                            | <a href="#">PH26</a>                                                                                        |
| Theorem 6.7: Convergence                                          | <a href="#">L43</a> , <a href="#">L44</a> , <a href="#">L47</a> , <a href="#">L55</a>                       |
| Theorem 4.2: Maximal Structural Rank in the Hard Family           | <a href="#">L64</a>                                                                                         |
| Theorem 3.4: Minimum Physical Bit Operations                      | <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">L43</a> , <a href="#">L46</a>                       |
| Theorem 4.3: Cutoff-Local Structural-Rank Bound for Exact Docking | <a href="#">L66</a>                                                                                         |
| Theorem 3.5: Decision-Class Bound                                 | <a href="#">IT4</a> , <a href="#">L43</a>                                                                   |
| Theorem 2.8: One Transition, One Bit                              | <a href="#">BA4</a>                                                                                         |
| Theorem 6.3: Rank Identification                                  | <a href="#">L43</a> , <a href="#">L46</a> , <a href="#">L51</a>                                             |
| Theorem 5.2: Rank-One Ground State                                | <a href="#">BA8</a> , <a href="#">L54</a> , <a href="#">L55</a>                                             |
| Theorem 2.9: Resolution Requires a Sufficient Coordinate Set      | <a href="#">BA5</a>                                                                                         |
| Theorem 4.5: Sampled Exact-Coarse Winner Preservation             | <a href="#">DQ2</a>                                                                                         |
| Theorem 4.6: Inside-Cutoff Sufficiency for Sampled Docking        | <a href="#">DQ1</a>                                                                                         |
| Theorem 6.5: Thermodynamic Selection                              | <a href="#">BA8</a> , <a href="#">L49</a> , <a href="#">L54</a> , <a href="#">L55</a>                       |
| Theorem 5.5: Exact-Resolution Time Lower Bound                    | <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">L43</a> |
| Theorem 6.4: Tractable Sufficiency at Rank One                    | <a href="#">L47</a> , <a href="#">L53</a> , <a href="#">L56</a>                                             |

Auto summary: mapped 54/54 (full=54, derived=0, unmapped=0).

## A.2 Lean Handle Map

| ID  | Lean Handle / Source                                                                                                   | ID   | Lean Handle / Source                                                                                                       |
|-----|------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------|
| AC1 | ClaimClosure.AtomicCircuitExports.AC1<br>paper4/DecisionQuotient/ClaimClosure.lean                                     | AC3  | ClaimClosure.AtomicCircuitExports.AC3<br>paper4/DecisionQuotient/ClaimClosure.lean                                         |
| AC4 | ClaimClosure.AtomicCircuitExports.AC4<br>paper4/DecisionQuotient/ClaimClosure.lean                                     | BA1  | Physics.BoundedAcquisition.BoundedRegion<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean                    |
| BA2 | Physics.BoundedAcquisition.acquisition_rate_<br>bound<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean   | BA3  | Physics.BoundedAcquisition.acquisitions_are_<br>transitions<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean |
| BA4 | Physics.BoundedAcquisition.one_bit_per_<br>transition<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean   | BA5  | Physics.BoundedAcquisition.resolution_reads_<br>sufficient<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean  |
| BA6 | Physics.BoundedAcquisition.srank_le_<br>resolution_bits<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean | BA7  | Physics.BoundedAcquisition.energy_ge_srank_<br>cost<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean         |
| BA8 | Physics.BoundedAcquisition.srank_one_energy_<br>minimum<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean | BA10 | Physics.BoundedAcquisition.counting_gap_<br>theorem<br>paper4/DecisionQuotient/Physics/<br>BoundedAcquisition.lean         |

(continued...)

| ID   | Lean Handle / Source                                                                                                                                               | ID   | Lean Handle / Source                                                                                                                                 |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| CV7  | Physics.Conversation.clamp_projection_eq_iff_same_clamped_bit<br>paper4/DecisionQuotient/Physics/Conversation.lean                                                 | CV8  | Physics.Conversation.clampDecisionEvent_iff_bitOps_pos<br>paper4/DecisionQuotient/Physics/Conversation.lean                                          |
| CV9  | Physics.Conversation.clamp_event_implies_positive_energy<br>paper4/DecisionQuotient/Physics/Conversation.lean                                                      | DQ1  | DecisionQuotient.Tractability.SampledDockingCutoff.sampled_insideCutoff_sufficient<br>paper4/DecisionQuotient/Tractability/SampledDockingCutoff.lean |
| DQ2  | DecisionQuotient.Tractability.SampledDockingGap.SampledDockingProblem.exact_coarse_opt_agree_of_gap<br>paper4/DecisionQuotient/Tractability/SampledDockingGap.lean | DT22 | Physics.DecisionTime.substrate_step_realizes_decision_event<br>paper4/DecisionQuotient/Physics/DecisionTime.lean                                     |
| DT23 | Physics.DecisionTime.substrate_step_is_time_unit<br>paper4/DecisionQuotient/Physics/DecisionTime.lean                                                              | DT24 | Physics.DecisionTime.time_unit_law_substrate_invariant<br>paper4/DecisionQuotient/Physics/DecisionTime.lean                                          |
| EI1  | ThermodynamicLift.energy_ge_kbt_nat_entropy<br>paper4/DecisionQuotient/ThermodynamicLift.lean                                                                      | IT3  | DecisionQuotient.quotientEntropy_le_srank_binary<br>paper4/DecisionQuotient/Information.lean                                                         |
| IT4  | DecisionQuotient.numOptClasses_le_pow_srank_binary<br>paper4/DecisionQuotient/Information.lean                                                                     | L17  | Leverage.compose_dof<br>Leverage/Foundations.lean                                                                                                    |
| L19  | Leverage.composition_dof_additive<br>Leverage/Theorems.lean                                                                                                        | L43  | dof_eq_srank<br>Leverage/BridgeToDQ.lean                                                                                                             |
| L44  | dof_one_iff_max_leverage<br>Leverage/Foundations.lean                                                                                                              | L45  | england_replication_inequality<br>Leverage/BridgeToDQ.lean                                                                                           |
| L46  | incoherent_srank_gt_one<br>Leverage/BridgeToDQ.lean                                                                                                                | L47  | max_coherence_forces_tractability<br>Leverage/BridgeToDQ.lean                                                                                        |
| L49  | srank_energy_lower_bound<br>Leverage/BridgeToDQ.lean                                                                                                               | L51  | ssot_srank_one<br>Leverage/BridgeToDQ.lean                                                                                                           |
| L52  | succ_le_two_pow<br>Leverage/BridgeToDQ.lean                                                                                                                        | L53  | sufficiency_conp_hard                                                                                                                                |
| L54  | thermodynamic_selection<br>Leverage/BridgeToDQ.lean                                                                                                                | L55  | thermodynamic_selection_unconditional<br>Leverage/BridgeToDQ.lean                                                                                    |
| L56  | tractable_bounded_core<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                                | L57  | Leverage.ColumnComplexityBridge.SharedCodewordCount_eq_TieBrokenRelationMoment<br>Leverage/ColumnComplexityBridge.lean                               |
| L58  | Leverage.ColumnComplexityBridge.zeroIdentityDebt_tieBrokenArgmin_of_uniform_argmin_relation_bound<br>Leverage/ColumnComplexityBridge.lean                          | L59  | Leverage.constrainedMolecular_energy_lower_bound<br>Leverage/BridgeToDQ.lean                                                                         |
| L60  | Leverage.rattle_constraintObservations_card<br>Leverage/BridgeToDQ.lean                                                                                            | L61  | Leverage.rattle_energy_lower_bound<br>Leverage/BridgeToDQ.lean                                                                                       |
| L62  | Leverage.rattle_srank_eq_effectiveDOF<br>Leverage/BridgeToDQ.lean                                                                                                  | L63  | Leverage.exactSufficiency_conp_core<br>Leverage/DockingTheoryBridge.lean                                                                             |
| L64  | Leverage.exactSufficiency_hardFamily_srank_eq_n<br>Leverage/DockingTheoryBridge.lean                                                                               | L65  | Leverage.molecularDocking_boundedPocket_srank_bound<br>Leverage/DockingTheoryBridge.lean                                                             |
| L66  | Leverage.molecularDocking_srank_bound<br>Leverage/DockingTheoryBridge.lean                                                                                         | ORA1 | oracle_arbitrary<br>paper2/Ssot/Coherence.lean                                                                                                       |
| PH26 | PhysicalComplexity.no_collapse_of_bounded_budget_pos_cost_exp_lb<br>paper4/DecisionQuotient/Physics/PhysicalHardness.lean                                          | QT1  | DecisionProblem.quotient_is_coarsest<br>paper4/DecisionQuotient/Quotient.lean                                                                        |
| QT2  | DecisionProblem.quotientMap_preservesOpt<br>paper4/DecisionQuotient/Quotient.lean                                                                                  | QT3  | DecisionProblem.quotient_represents_opt_equiv<br>paper4/DecisionQuotient/Quotient.lean                                                               |
| QT7  | DecisionProblem.quotient_has_unique_factorization<br>paper4/DecisionQuotient/Quotient.lean                                                                         | SE1  | ClaimClosure.SE1<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                        |

(continued...)

| ID   | Lean Handle / Source                                                                                                                                                          | ID   | Lean Handle / Source                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| SE2  | ClaimClosure.SE2<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                                                 | SE3  | ClaimClosure.SE3<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                          |
| SE4  | ClaimClosure.SE4<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                                                 | SE5  | ClaimClosure.SE5<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                          |
| SE6  | ClaimClosure.SE6<br>paper4/DecisionQuotient/ClaimClosure.lean                                                                                                                 | W1   | Physics.single_future_zero_cost<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean                                                       |
| W2   | Physics.transportCost_pos_of_offDiag<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean                                                                         | W3   | Physics.integrity_is_centroid<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean                                                         |
| W4   | Physics.wasserstein_bridge<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean                                                                                   | WM4  | Physics.WolpertMismatch.mismatchNatLowerBound_<br>pos_of_exists_ne<br>paper4/DecisionQuotient/Physics/<br>WolpertMismatch.lean                         |
| WM6  | Physics.WolpertDecomposition.effective_model_<br>strictly_exceeds_landauer_of_distribution_<br>mismatch<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean      | WP2  | Physics.WolpertDecomposition.landauer_floor_<br>plus_decomposition_lower_bound<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean        |
| WP6  | Physics.WolpertDecomposition.effective_model_<br>strictly_exceeds_landauer_of_either_cited_<br>component<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean     | WP7  | Physics.WolpertDecomposition.landauer_floor_<br>plus_structural_resource_lower_bound<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean  |
| WP8  | Physics.WolpertDecomposition.energy_lower_<br>bound_increases_by_structural_resource<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean                         | WP9  | Physics.WolpertDecomposition.physical_<br>grounding_bundle_with_wolpert_decomposition<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean |
| WR6  | Physics.WolpertResidual.<br>discreteResidualNatLowerBound_pos_of_<br>asymmetry_or_oneway<br>paper4/DecisionQuotient/Physics/<br>WolpertResidual.lean                          | WR7  | Physics.WolpertDecomposition.stopping_time_<br>residual_of_discrete_edge_split<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean        |
| WR10 | Physics.WolpertDecomposition.effective_model_<br>strictly_exceeds_landauer_of_finite_discrete_<br>witness<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean    | WR11 | Physics.WolpertResidual.<br>binaryEncodedResidualNatLowerBound_eq_one<br>paper4/DecisionQuotient/Physics/<br>WolpertResidual.lean                      |
| WR12 | Physics.WolpertDecomposition.effective_<br>model_ge_landauer_plus_one_of_binary_encoded_<br>residual_example<br>paper4/DecisionQuotient/Physics/<br>WolpertDecomposition.lean |      |                                                                                                                                                        |

### A.3 Proof Hardness Index

| Paper handle                 | Hardness profile | Regime tags | Lean support                                                                                                                                                                  |
|------------------------------|------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cor:bounded-pocket-regime    | unspecified      | -           | <a href="#">L65</a>                                                                                                                                                           |
| cor:                         | unspecified      | -           | <a href="#">IT3</a> , <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">EI1</a> , <a href="#">L43</a>                       |
| budget-entropy-bound         | unspecified      | -           | <a href="#">L17</a> , <a href="#">L19</a> , <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA7</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">L43</a> |
| cor:holonomic-landauer-floor | unspecified      | -           | <a href="#">L60</a> , <a href="#">L61</a> , <a href="#">L62</a>                                                                                                               |
| cor:                         | unspecified      | -           | <a href="#">BA8</a> , <a href="#">L54</a> , <a href="#">L55</a>                                                                                                               |
| minimum-cost-regime          | unspecified      | -           | <a href="#">L46</a> , <a href="#">L51</a>                                                                                                                                     |
| cor:rank-above-one           | unspecified      | -           | <a href="#">L51</a>                                                                                                                                                           |
| cor:rank-one                 | unspecified      | -           | <a href="#">AC1</a> , <a href="#">AC3</a> , <a href="#">AC4</a>                                                                                                               |
| prop:                        | unspecified      | -           |                                                                                                                                                                               |
| atomic-realization           |                  |             |                                                                                                                                                                               |

|                                    |             |   |                                          |
|------------------------------------|-------------|---|------------------------------------------|
| prop:                              | unspecified | - | IT3, BA5, BA6, WP2, WM4, L43             |
| binary-mismatch-cumulative-work    |             |   |                                          |
| prop:                              | unspecified | - | IT3, BA5, BA6, WP2, WM4, L43             |
| binary-mismatch-energy-information |             |   |                                          |
| prop:binary-mismatch-example       | unspecified | - | BA5, BA6, WP2, WM4, L43                  |
| prop:                              | unspecified | - | IT3, BA5, BA6, WR12, WR11, L43           |
| binary-residual-cumulative-work    |             |   |                                          |
| prop:binary-residual-example       | unspecified | - | IT3, BA5, BA6, WR12, WR11, L43           |
| prop:bounded-region                | unspecified | - | BA1                                      |
| prop:canonical-wolpert-bundle      | unspecified | - | BA5, BA6, WP9, L43                       |
| prop:dof-additive                  | unspecified | - | L17, L19                                 |
| prop:ei-hierarchy                  | unspecified | - | IT3, BA5, BA6, WR12, WP2, WM4, WR11, L43 |
| prop:finite-compression-bridge     | unspecified | - | L57, L58                                 |
| prop:finite-discrete-residual      | unspecified | - | WR10, WR7, WR6                           |
| prop:finite-lifetime               | unspecified | - | SE1, SE2, SE3, SE4, SE5                  |
| prop:                              | unspecified | - | SE5, IT3, L43                            |
| lifetime-throughput                |             |   |                                          |
| prop:                              | unspecified | - | QT2, QT7, QT1, QT3                       |
| optimizer-quotient                 |             |   |                                          |
| prop:                              | unspecified | - | SE6                                      |
| speed-heat-tradeoff                |             |   |                                          |
| prop:strict-canonical-energy       | unspecified | - | BA5, BA6, WP6, L43                       |
| prop:strict-overhead               | unspecified | - | BA5, BA6, WM6, WP6, WR10, L43            |
| prop:structural-resource-overhead  | unspecified | - | BA5, BA6, WP8, WP7, L43                  |
| prop:                              | unspecified | - | DT23, DT22, DT24                         |
| substrate-time-law                 |             |   |                                          |
| prop:                              | unspecified | - | CV8, CV9, CV7                            |
| threshold-channel                  |             |   |                                          |
| thm:                               | unspecified | - | BA1, BA2                                 |
| bounded-acquisition                |             |   |                                          |
| thm:                               | unspecified | - | IT4, IT3, BA1, BA2, BA5, BA6, EI1, L43   |
| budget-class-bound                 |             |   |                                          |
| thm:coherent-single-source         | unspecified | - | ORA1                                     |
| thm:counting-gap                   | unspecified | - | BA10                                     |
| thm:                               | unspecified | - | BA3                                      |
| discrete-acquisition               |             |   |                                          |
| thm:dof-srank                      | unspecified | - | L43                                      |
| thm:energy-entropy                 | unspecified | - | IT3, EI1, L43                            |
| thm:energy-rank                    | unspecified | - | BA7, BA6, L43                            |
| thm:england                        | unspecified | - | L45                                      |
| thm:entropy-bound                  | unspecified | - | IT3, L43                                 |
| thm:                               | unspecified | - | L63                                      |
| exact-sufficiency-hardness-core    |             |   |                                          |
| thm:finite-budget-no-collapse      | unspecified | - | PH26                                     |
| thm:five-way                       | unspecified | - | L43, L44, L47, L55                       |
| thm:                               | unspecified | - | L64                                      |
| hard-family-srank                  |             |   |                                          |
| thm:                               | unspecified | - | BA5, BA6, L43, L46                       |
| min-bit-operations                 |             |   |                                          |
| thm:molecular-docking-srank-bound  | unspecified | - | L66                                      |
| thm:numopt-bound                   | unspecified | - | IT4, L43                                 |
| thm:one-transition-one-bit         | unspecified | - | BA4                                      |

|                                      |             |   |                                                                                                             |
|--------------------------------------|-------------|---|-------------------------------------------------------------------------------------------------------------|
| thm:                                 | unspecified | - | <a href="#">L43</a> , <a href="#">L46</a> , <a href="#">L51</a>                                             |
| rank-identification                  |             |   |                                                                                                             |
| thm:rank-one-ground                  | unspecified | - | <a href="#">BA8</a> , <a href="#">L54</a> , <a href="#">L55</a>                                             |
| thm:resolution-sufficient            | unspecified | - | <a href="#">BA5</a>                                                                                         |
| thm:                                 | unspecified | - | <a href="#">DQ2</a>                                                                                         |
| sampled-docking-gap                  |             |   |                                                                                                             |
| thm:sampled-inside-cutoff-sufficient | unspecified | - | <a href="#">DQ1</a>                                                                                         |
| thm:thermodynamic-selection          | unspecified | - | <a href="#">BA8</a> , <a href="#">L49</a> , <a href="#">L54</a> , <a href="#">L55</a>                       |
| thm:time-lower-bound                 | unspecified | - | <a href="#">BA1</a> , <a href="#">BA2</a> , <a href="#">BA5</a> , <a href="#">BA6</a> , <a href="#">L43</a> |
| thm:                                 | unspecified | - | <a href="#">L47</a> , <a href="#">L53</a> , <a href="#">L56</a>                                             |
| tractable-rank-one                   |             |   |                                                                                                             |

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*Auto summary: indexed 54 claims by hardness profile (unspecified=54).*

## B Scope Statements

This appendix lists the principal scope statements for the finite decision-thermodynamic framework:

- **Canonical encoding:** The main theorems are exact for the canonical binary decision problem attached to the bounded decision system. Other physical encodings require an explicit transport theorem.
- **Structural chain:** Counting Gap fixes the finite-event statement. Bounded Acquisition fixes the traversal-rate statement. Discrete Acquisition and One Transition, One Bit fix the acquisition-event interface.
- **Landauer calibration:** Thermodynamic cost is calibrated by a per-bit Landauer floor. Stronger substrate-dependent lower bounds may exist, but they are additional assumptions, not part of the theorem package under discussion.
- **Exact-resolution setting:** The results concern exact sufficiency and exact-resolution cost. Approximate, stochastic, or bounded-confidence regimes require separate analysis.
- **Finite state family:** The entropy and replication theorems are finite counting results. Continuum models must first be reduced to a finite decision quotient before these arguments apply.
- **Replication theorem:** The finite replication entropy gap is a theorem of the calibrated exact-resolution model. England’s 2013 theorem belongs to a stochastic-thermodynamic path-space model.
- **Finite-budget theorem:** Finite-Budget No-Collapse is a theorem about bounded budgets, positive per-bit cost, and exponential lower-bound growth.

Remark B.1 (Scope Statement: Molecular Independence Hypothesis). Corollary [5.26](#) proves the finite constrained-molecular transport once the RATTLE holonomic topology supplies  $k$  independent constraints and the corresponding binary status interface. It does not yet derive that independence hypothesis from a concrete geometric constraint family and molecular topology object. Establishing that derivation is additional work beyond the present theorem package, not a hidden premise of the corollary.

## C Complete Theorem Index

Paper-level labeled claims:

### **Foundations (Section 2):**

- Proposition [2.3](#) (DOF Additivity)
- Theorem [2.4](#)
- Proposition [2.5](#)
- Theorem [2.6](#)
- Theorem [2.7](#)
- Theorem [2.8](#)
- Theorem [2.9](#)

### **Structural Rank and Decision Entropy (Section 3):**

- Theorem [3.1](#)
- Corollary [3.2](#)
- Corollary [3.3](#)
- Theorem [3.4](#)
- Theorem [3.5](#)
- Theorem [3.6](#)
- Proposition [3.7](#)

### **Complexity Boundary (Section 4):**

- Theorem [4.1](#)
- Theorem [4.2](#)
- Theorem [4.3](#)
- Corollary [4.4](#)
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### **Thermodynamic Cost (Section 5):**

- Theorem [5.1](#)
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- [Theorem 5.5](#)
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- [Proposition 5.21](#)
- [Proposition 5.22](#)
- [Proposition 5.23](#)
- [Proposition 5.24](#)
- [Proposition 5.25](#)
- [Corollary 5.26](#)

#### **Convergence and Universal Consequences (Section 6):**

- [Theorem 6.7](#)
- [Theorem 6.8](#)
- [Theorem 6.9](#)

#### **Primary Lean sources:**

- `Leverage/Foundations.lean`
- `Leverage/BridgeToDQ.lean`

- `Leverage/ColumnComplexityBridge.lean`
- `Leverage/DockingTheoryBridge.lean`
- `LambdaDR.lean`
- `Leverage.lean`

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